Taylor-series item review

My role Math SME and item review

I reviewed higher-ed calculus items on binomial and Taylor series, including learning objectives and answer choices. The excerpt shows the section structure and the questions under review.

ORIGINAL WORK FILE EXCERPT

Chapter 10: Power Series

Section 10.4 Working with Taylor Series

Binomial Series

Learning Objectives

Use a binomial series

Functions Expressed as Taylor Series

Learning Objectives

Find the Taylor expansion of common functions

Find the Taylor expansion of common functions

None of these.

Find the radius of convergence in a Taylor Series

The function f(x) = is a Taylor series centered at . Which of the following is the correct expression for the Taylor series and radius of convergence?